



Hanging by the Thread: An Empirical Study of Donations in Open Source

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1 INTRODUCTION AND MOTIVATION

Open source plays a critical role in our software infrastructure. It is used in the creation of almost every product and makes it increasingly easy to create powerful software cheaply and quickly, which many companies benefit from. However, its importance and our dependence on it, are often not recognized [2]. Like all software projects, open source needs maintenance to fix bugs and adapt code to evolving technologies [12]. With increasing popularity, demands for maintenance and support work also rise, resulting in many requests and reported issues. How to supply all of the needed maintenance and development work is an open and sometimes controversial question.

There are many different approaches to fund the creation and maintenance of open source [3], including selling services around a project, adopting a combined open/closed source strategy (e.g., dual license, ‘open core’), or just asking for donations. In many cases, companies and open-source projects band together and create foundations to support open-source projects [10]. Yet, much of current open-source work seems to be still performed by volunteers [23]. Developers who volunteer their time for open source increasingly report stress and even burnout [e.g., 6, 11, 20], and turnover can be high, threatening the sustainability of critical projects [1, 4, 8, 9, 13, 15, 16, 19, 21, 22]. Some developers thus hope that donations are a path to sustain their own open-source activities.

While donations are often discussed by practitioners [e.g., 7, 17, 18], little is known about their prevalence, success, and impact. Through an exploratory mixed-methods study, we research the goals, reality, and use of donation and crowd-funding platforms, such as *OpenCollective*, *PayPal*, and *Patreon*, in open-source software projects, contributing a deeper understanding of this new phenomenon where there was previously only controversial discussion.

We (1) perform a census of crowd-funding platforms used by open-source projects hosted on GITHUB to understand how common the phenomenon is, (2) statistically model the characteristics of projects asking for and receiving donations, (3) qualitatively analyze publicly available artifacts to identify what the requested funds are needed for, (4) use time series analysis to assess whether the received donations achieve the intended outcomes, and (5) explore how received donations are spent.

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ICSE '20 Companion, May 23–29, 2020, Seoul, Republic of Korea

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ACM ISBN 978-1-4503-7122-3/20/05...\$15.00

<https://doi.org/10.1145/3377812.3382170>

2 RESEARCH METHODS

Our primary strategy to identify which repositories ask for donations and with which donation platforms was to analyze README.md files of GITHUB repositories for links or mentions of specific donation platforms. These repositories are part of a larger GITHUB corpus of 77,934,441 repositories, excluding forks, indexed by GHTORRENT [5] as of May 23, 2019, the date GITHUB announced its own donation platform feature [24].

Data mining. We iteratively curated a list of 12 donation platforms and corresponding search strings that would identify them in README.md markdown files. The platforms include: *Bountysource*, *Flattr*, *IssueHunt*, *Kickstarter*, *Liberapay* (including *Gittip* and *Gratipay*), *OpenCollective*, *Otechie*, *Patreon*, *PayPal*, *SALT*, *Tidelift*, and *Tip4Commit*. We relied on the GITHUB API to search for projects that contained the donation-service-specific terms in their README.mds. We found 25,885 projects asking for donations and identified when each donation platform was first introduced from the git history of the README.md files. We also collected a number of characteristics that capture different dimensions of activity and popularity, including the *total number of commits* up until May 23rd, a binary flag *is active* indicating whether the repository had at least one commit in the 9 months before that day, the *total number of issues* on GITHUB, and the *total number of stars* on GITHUB. For longitudinal analyses, we computed monthly commits, donations, issues closed, and issue resolution speed.

Data analysis. We used a combination of qualitative and quantitative methods to investigate the emerging and largely unexplored phenomenon of donations in open source.

Multiple regression modeling. We implemented multiple regression modeling to identify characteristics of projects that are more likely to ask for donations and of those that are successful in raising donations.

Qualitative text analysis. To understand developer expectations for donations, we qualitatively analyzed the text of README.md files and donation profile pages in 109 repositories using *card sorting* [14]. We also used qualitative analysis to explore the difference between expectations and outcomes, focusing on how donations were used in 60 projects.

Longitudinal analysis. We modeled the longitudinal effects of donations on a sample of 337 projects that successfully raised money using interrupted time series design with the date of the first donation received via *Patreon* or *OpenCollective* as the intervention. We considered projects with at least 9 months of activity before and after their first donations.

3 RESULTS

Frequency of donation requests. There are many projects (25,885) asking for donations, but they make up only 0.04% of our corpus. As summarized in the left part of Figure 1, *PayPal* is the most

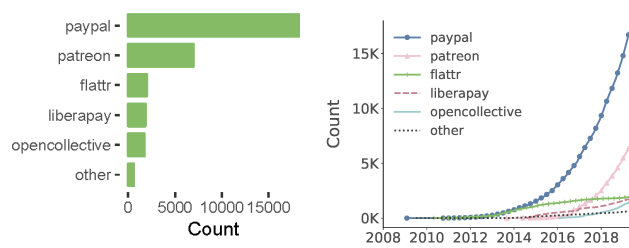


Figure 1: Most popular donation services across GITHUB (left) and the adoption of donation platforms over time on GITHUB (right).

common donation platform by number of repositories in GITHUB, followed by *Patreon*, *Flattr*, *Liberapay*, and *OpenCollective* have similar counts, then, with much lower adoptions, come other services.

Developers slowly started to ask for donations (in the form of links to donation platforms) around 2012, with a significant increase in recent years, as visible in the right part of Figure 1; the only stagnating platform is *Flattr*, likely related to controversy around its sale in 2017.

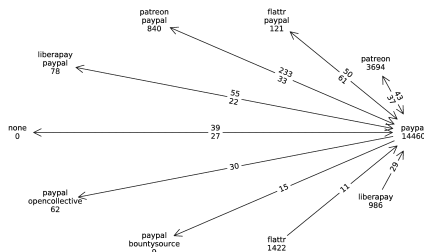


Figure 2: Transitions between donation platforms involving *PayPal*, the numbers below platform labels indicate the number of repositories that never transitioned. For edges with two labels, the top number corresponds to transitions from *PayPal* while the bottom number corresponds to transitions to *PayPal*. All edges with weights less than 10 are excluded.

While most repositories adopt only a single donation platform, 10.9 % of repositories ask for donations with multiple platforms. (*PayPal* + *Patreon* is the most common combination.) As depicted in Figure 2, transitions between donation platforms are relatively rare after the initial adoption, so is the removal of donation links. The highest number of transitions, 233, occurs when repositories with *PayPal* add *Patreon* to their README .mds.

Project characteristics. The projects asking for and receiving donations tend to be more active, more mature, and more popular. However, we observe a highly skewed distribution of received funds: 58 % of repositories received donations, and only 10 % received 1000 USD/month or more.

Expectations for donations. Donations are requested predominantly, but not exclusively, to support engineering activities. From card sorting, we identify four categories of donation requests: engineering, community, project expenses, and personal.

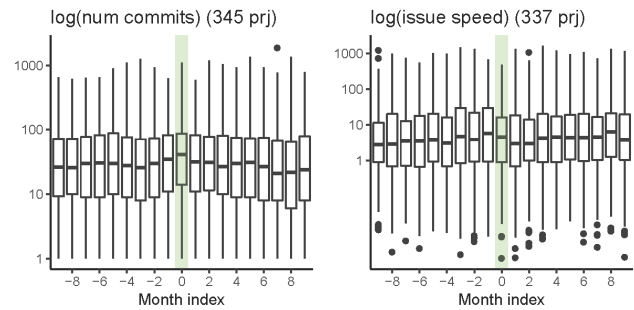


Figure 3: The distribution of outcomes before and after receiving the first donation.

Observable outcomes of donations. Figure 3 shows how the distributions of two monthly outcome measures, *number of commits* and *issue resolution speed*, evolve before and after projects start receiving donations. Visually, the trends appear stationary. From the interrupted time series regression models, we find some evidence, but not strong support, for the hypothesis that donations lead to higher levels of development or maintenance activity. We also find that projects asking for but not receiving any donations actually decline in activity.

Use of donations. Exploring this seeming mismatch between expectations and outcomes, we study how raised funds are spent and find that they are often spent in line with purposes for which they were requested (if sufficient funds are raised), but also that many projects actually only spend a fraction of their funds and that funding contributors beyond the top contributors is not uncommon.

4 CONTRIBUTIONS

We found a small percentage of projects asking for donations on GITHUB, most commonly with *PayPal* and *Patreon*, often to support engineering activities, but also for community services, web hosting, and personal expenses. Many of these projects receive donations, but rarely enough to fund a full time engineering position. While we did not find strong evidence that received donations associate with higher levels of activity in a project, we observed different patterns of how received donations are spent.

Our study is exploratory in nature and can only be a first step in understanding the current state of donations in open source using public data. Overall, we contribute a design to identify projects requesting open source donations, a census of donation requests on GITHUB, an analysis of characteristics of projects requesting and receiving donations, a list of commonly stated reasons for asking for donations, a time-series analysis of donations' effects on project activity, and an analysis of how donations are commonly spent. Our work reveals insights that can be starting points for many interesting future research directions, including *What level of funding is actually needed to sustain open-source projects?* and *What kind of spending is effective at sustaining open-source projects?*

While we expect that donations will always be only one of many mechanisms in sustaining open source overall, we believe it may be an important one, given that it is easy to adopt (compared to

founding a company), may support a large range of activities (including community support and travel), is broadly supported (now also directly by GITHUB), and seems to have some acceptance from both individuals and corporations.

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